

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2023-2024
M.Tech. in AIML

Mid-Semester Test
(EC-2 Regular Paper)

Course No. : AIMLCZG530
Course Title : Natural Language Processing
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 21-01-2024_FN

No. of Pages	= 3
No. of Questions	= 5

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Question 1. [3+2+2=7 Marks]

A. Find the ambiguity of the below sentences and justify your answer [3 marks]

a) The tank is full of water. I saw a military tank.

Soln:

Semantic ambiguity

b) Before the professor left the stage, the play begins

Syntactic ambiguity

c) She is looking for a match

Lexical ambiguity

B.

i. Given is the following toy corpus. Calculate all the bigram probabilities. [2 marks]

<s> I like apples </s>

<s> Apple is good for health </s>

<s> Apple is in red colour </s>

Soln:

Handwritten solution for bigram probabilities:

<s> I like apples </s>
<s> Apple is good for health </s>
<s> Apple is in red colour </s>

$$P(I | <s>) = \frac{1}{3} = 0.33 \quad P(\text{like} | I) = \frac{1}{1} = 1$$
$$P(\text{apples} | \text{like}) = \frac{1}{1} = 1 \quad P(\text{is} | \text{apples}) = \frac{1}{2}$$
$$P(\text{apple} | <s>) = \frac{2}{3} = 0.67 \quad P(\text{is} | \text{apple}) = \frac{1}{2}$$
$$P(\text{good} | \text{is}) = \frac{1}{2} = 0.5 \quad P(\text{for} | \text{good}) = \frac{1}{1} = 1$$
$$P(\text{health} | \text{for}) = \frac{1}{1} = 1 \quad P(\text{is} | \text{health}) = \frac{1}{1} = 1$$
$$P(\text{apple} | \text{is}) = \frac{1}{1} = 1$$
$$P(\text{in} | \text{is}) = \frac{1}{1} = 1 \quad P(\text{red} | \text{in}) = \frac{1}{1} = 1$$
$$P(\text{colour} | \text{red}) = \frac{1}{1} = 1 \quad P(\text{is} | \text{colour}) = \frac{1}{1} = 1$$

- ii. For above training data in (i), Calculate the probability of below sentence using raw bigram probabilities and using Laplace smoothing, <s> I am eating apples</s> [2marks]

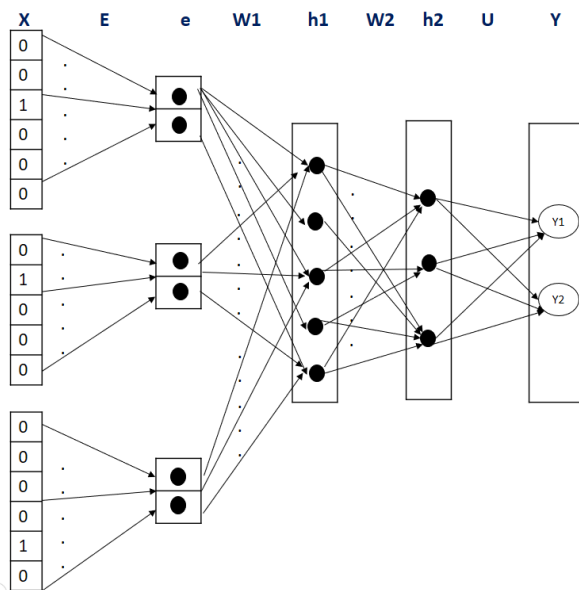
$$\begin{aligned}
 P(I|<s>) &= 0.33 \\
 P(am|I) &= 0 \\
 P(eating|am) &= 0 \\
 P(apples|eating) &= 0 \\
 P(</s>|apples) &= \frac{1}{1} = 1 \\
 0.33 \times 0 \times 0 \times 0 \times 1 &= 0
 \end{aligned}$$

Using Laplace smoothing

$$\begin{aligned}
 P(I|<s>) &= \frac{1+1}{3+1} = \frac{2}{4} = 0.5 \\
 P(am|I) &= \frac{1}{11} = 0.0909 \\
 P(eating|am) &= \frac{1}{11} = 0.0909 \\
 P(apples|eating) &= \frac{1}{11} = 0.0909 \\
 P(</s>|apples) &= \frac{1+1}{1+11} = \frac{2}{12} = \frac{1}{6} = 0.1667 \\
 &= 0.000163296
 \end{aligned}$$

Question 2. [4+3 = 7 Marks]

- A. Study the below neural network designed for learning word embedding along with other NLP application stated below and answer the following questions.



If the input layer 'X' denote the one hot encoding of the vocabulary, "e" is the embedding layer, "h1", "h2" are hidden layers and "Y" is the output layer emitting continuous valued output, identify no more than 3 issues/error in the architecture and suggest modification to suit the below use case requirement. If there are no corrections required, then mention "No Error" explicitly.

Use Case: Neural network in language modelling for sentence completion

Given a training corpus with below vocabulary each vectorized with four dimensions, and following test sentence phrase, the neural network, should have predictive ability to identify next best word to fill in the blank of the test sentence, by analyzing context window with five tokens.

Vocabulary: {he, she, bat, tree, wooden, park, playing, saw, was, a, on, the, with, in, morning, evening} Test Sentence: "on a morning he was playing in the _____"

Solution & Marking Scheme:

1 Mark: Each for below error & suggestions:

- No.of.one-hot-encoding vectors must be 5 but is three in number
- Output needs $|V|=16$ no.of.neuron for Y-Pred
- One-hot-encoding size is not consistent ☹️Change to size of vocabulary = 16
- Embedding size 'E' should be 4 in dimension but its two

B. The number of times each word appears in different documents is given in the table below.

Calculate the TF-IDF value for each term in D1. **[1.5 mark]**

Find the word embedding for each term using TF-IDF value. Find which words are closest using TF-IDF word embedding. **[0.5 mark]**

Which documents are more similar to each other? **[0.5 mark]**

What is the disadvantage of using the TF-IDF values for the word embedding's? **[0.5 mark]**

	D1	D2	D3
NLP	10	0	0
is	50	66	89
extremely	20	22	12
interestin g	30	32	11
course	20	0	0

Solution

	D1	N	df	idf
NLP	10	3	1	0.477
tf	1.04			
tf-idf	0.496			
is	50	3	3	0
tf	1.70			
tf-idf	0			
extremely	20	3	3	0
tf	1.32			
tf-idf	0			
interesting	30	3	3	0
tf	1.49			
tf-idf	0			
course	20	3	1	0.477
tf	1.32			
tf-idf	0.63			

Find the word embedding for each term using TF-IDF value.

NLP [0.496,0,0]

is [0,0,0]

extremely [0,0,0]

interesting[0,0,0]

course [0.63,0,0]

Find which words are closest using TF-IDF word embedding. **[0.5 mark]**

Which documents are more similar to each other? **[0.5 mark] D2 and D3**

What is the disadvantage of using the TF-IDF values for the word embedding's? [0.5 mark] - Sparsity

Question 3. [1.5+3+1.5=6 Marks]

Given a training corpus: "played in the morning", use the skip-gram negative sampling method and answer the following: The initial embedding matrix and initial context matrix has dimensions $|v| \times 3$ and is given as follows:

Note: No need to update or show any weights other than necessary for below questions. Follow only the approach as discussed in class. i.e., Simplified Skip gram negative sampling with binary classification model. Round all the calculations to exactly two decimal places.

Vocabulary	Initial Embedding Matrix		
on	1	0	0
in	0	1	1
morning	1	0	0
he	1	1	0
was	0	1	1
played	1	0	0
with	0	1	1
the	0	0	1
bat	1	0	0

Vocabulary	Initial Context Matrix		
on	0	1	0
in	0	0	1
morning	0	1	0
he	1	0	1
was	0	0	1
played	0	1	0
with	0	0	1
the	1	0	0
bat	1	1	0

- Generate the training dataset for an input target word "played" and context window of 1 next word and hyper parameter value $k=2$ for the negative sampling task. Use the information available in the question.
- Calculate the error for the above dataset for only the first iteration of skip-gram training, with only one hidden layer.
- Explain in no more than 40 words, why skipgram algorithm training was modified from multiclass to binary classification task.

Solution & Marking Scheme:

- a) Training data: 1.5 mark

Target	Context	Label
played	in	1
played	the	0
played	morning	0

- b) Error:

Dot product = 1 mark

Sigmoid = 1 mark

Error = 1 mark

	context embeddings			i/p word embeddin g	dot product	sigmoid()	t	sig-t
in	0	0	1	1	0	0.5	1	-0.5
the	1	0	0	0	1	0.73	0	0.73
morning	0	1	0	0	0	0.5	0	0.5

- c) 0.5 mark Faster/Ease of Computation.

1 mark: Instead of updating all weights, 1 iteration updates only the context word & target word's embedding for a training instance

Question 4. [5 Marks]

Find the appropriate POS tag using statistical model with bigram assumption, for the word "cook" in the sentence,

“He will cook the food”

Word to Tag combination		
1	p(cook/VB)	0.0056
2	p(cook/NN)	0.0072
3	p(the/DT)	0.014
4	P(food/NN)	0.00047

Tag to Tag combination		
1	p(VB/MD)	0.78
2	p(NN/MD)	0.14
3	p(DT/VB)	0.075
4	p(DT/NN)	0.026

Note:

NN- Noun

VB-Verb

MD-Modal

DT-Determiner

Solution:

$p(\text{VB/MD}) * p(\text{DT/VB}) * p(\text{cook/VB}) = 0.0003276$

$p(\text{NN/MD}) * p(\text{DT/NN}) * p(\text{cook/NN}) = 0.000026208$

The most appropriate POS tag is VB

Question 5. [5 Marks]

By using Viterbi Algorithm, fill the Viterbi table for the sentence, “He will fight”. The tag transition probabilities and the word likelihood for this corpus are as follows:

Tag transition probabilities	MD	NN	VB	PRP
MD	0.000008	0.31	0.46	0.0056
NN	0.000096	0.209	0.658	0.00068
VB	0.001	0.05	0	0.008
PRP	0.08	0.02	0.001	0.00001
START	0.008	0.000934	0.05677	0.08

Word likelihood probabilities	h e	will	fight
MD	0	0.8	0
NN	0	0.2	0.4
VB	0	0	0.6
PRP	1	0	0

Viterbi Table	H E	WILL	FIGHT
---------------	--------	------	-------

NN			
VB			
MD			
PRP			

Note:

PRP: PERSONAL PRONOUN
MD:MODAL
VB:VERB BASE FORM
NN:NOUN, SINGULAR OR
MASS

Solution

Viterbi Table	HE	WILL	FIGHT
NN	0	0.00032	0.00063488
VB	0	0	0.00141312
MD	0	0.00512	0
PRP	0.08		0